THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL CHAPTER TEST SET 3

Chapter 13: **Two Variables, One Line** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The line $kx + 3y = 0$ passes through $(1, -4)$. Its slope is: 1

(a) -5 **(b)** -2 **(c)** -3 **(d)** -4

2. The line $16x + 7y = 112$ has intercepts p on the x-axis and q on the y-axis. The value of $p + q$ is: 1

(a) 135 **(b)** 23 **(c)** 22 **(d)** 24

3. Using $F = \frac{9}{5}C + 32$, the temperature $115°$C in degrees Fahrenheit is: 1

(a) 238 **(b)** 248 **(c)** 240 **(d)** 239

4. The slope of the line through $(-7, 9)$ and $(17, -1)$ is: 1

(a) $-\frac{17}{12}$ **(b)** $\frac{19}{12}$ **(c)** $\frac{7}{12}$ **(d)** $-\frac{5}{12}$

5. The line $17x + 22y = 374$ cuts the axes at two points. The area of the triangle it forms with the axes (square units) is: 1

(a) 188 **(b)** 186 **(c)** 189 **(d)** 187

6. A taxi fare is ₹131 for 2 km and ₹347 for 8 km (fare is a linear function of distance). The fare for 35 km (in ₹) is: 1

(a) 1283 **(b)** 1355 **(c)** 1378 **(d)** 1319

7. The line $x + 7y = 9$ is parallel to the line $kx + 35y = 11$. The value of k is: 1

(a) 7 **(b)** 6 **(c)** 5 **(d)** 4

8. The points $(4, 3)$, $(5, 4)$ and $(6, k)$ are collinear. The value of k is: 1

(a) 6 **(b)** 5 **(c)** 7 **(d)** 4

9. **Assertion (A):** The slope of the line $3x + y = 7$ is -3. 1

Reason (R): Writing $3x + y = 7$ as $y = -3x + 7$ shows that the slope is the coefficient of x.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. **Assertion (A):** The line $y = 2x$ passes through the origin. 1

Reason (R): The slope of the line $y = 2x$ is 2.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the equation of the line through $(-13, -6)$ and $(4, 11)$ in the form $y = mx + c$. 2

12. The line $kx + 3y = 15$ passes through the point $(1, 1)$. Find k, the slope and the y-intercept of the line. 2

13. Find the area of the triangle formed by the line $21x + 24y = 504$ and the coordinate axes. 2

14. Find the equation of the line through $(-8, -25)$ and $(-1, 10)$ in the form $y = mx + c$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A line cuts the x-axis at $(9, 0)$ and the y-axis at $(0, 7)$. Find its equation, its slope and the area of the triangle it makes with the axes. 3

16. A water supplier charges a fixed amount plus a rate per kilolitre. For 6 kL the bill is ₹88 and for 12 kL it is ₹136. Find the equation of the bill, and the usage whose bill is ₹192. Also find the bill for no usage. 3

17. Find k if the points $(19, 15)$, $(21, 9)$ and $(23, k)$ lie on one straight line. 3

OR

A line cuts the x-axis at $(15, 0)$ and the y-axis at $(0, 13)$. Find its equation, its slope and the area of the triangle it makes with the axes. 3

18. A water supplier charges a fixed amount plus a rate per kilolitre. For 5 kL the bill is ₹125 and for 12 kL it is ₹181. Find the equation of the bill, and the usage whose bill is ₹237. Also find the bill for no usage. 3

OR

Find k if the points $(-20, -26)$, $(-18, -30)$ and $(-16, k)$ lie on one straight line. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** A rectangular field has a perimeter of 30 m. Its length is l m and its breadth is b m. 5

(i) Write a linear equation in l and b.

(ii) Find b when $l = 9$.

(iii) Is $(10, 6)$ a possible pair of length and breadth?

(iv) How many solutions in positive whole numbers are possible, and what are the intercepts of the line?

20. A line passes through $(-4, -8)$ and $(8, 4)$. Find its equation, then find k if the point $(k, 3)$ lies on it, and the x-intercept. 5

OR

A taxi fare is ₹313 for 5 km and ₹463 for 8 km. Assuming the fare is linear in the distance, find the fare per km, the fixed charge and the fare for 37 km. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND